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Fluidic group and related apparatus for drug preparation

Abstract

A fluidic group couplable to a first container adapted to contain a drug and a second container adapted to contain an ingredient, comprising: a syringe comprising a movable piston defining a chamber having a variable volume; a first and a second coupling group selectively fixable to the second and, respectively, to the first container; a forking element having a first, a second and a third inlet in fluid communication with each other and, respectively, with the first coupling group, with the second coupling group and with the chamber; a first one-way valve interposed between the first coupling group and the forking element and controllable to allow the passage of the ingredient from the second container to the chamber; and a second one-way valve interposed between the second coupling group and the forking element and controllable to allow the passage of the ingredient from the chamber to the first container.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10925808	12/2020	Pak	N/A	A61J 1/20
2015/0053305	12/2014	Davidian	141/2	A61J 1/22
2015/0087868	12/2014	Lee	422/159	C07B 59/00
2015/0283322	12/2014	Hachey et al.	N/A	N/A
2022/0023531	12/2021	McLoughlin	N/A	A61M 5/14248
2022/0362106	12/2021	Gyory	N/A	A61J 1/2037

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2007/122209	12/2006	WO	N/A
2014/204894	12/2013	WO	N/A
2015/071852	12/2014	WO	N/A
2017/096072	12/2016	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion corresponding to International Patent Application No. PCT/IB2021/053733, mailed Jul. 2, 2021. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is filed pursuant to 35 U.S.C. 371 as a U.S. National Phase application

of International Patent Application No. PCT/IB2021/053733, which was filed May 4, 2021, claiming priority from Italian patent application no. 102020000009751 filed on Apr. 5, 2020. The entire text of the aforementioned applications is incorporated herein by reference in its entirety.
TECHNICAL FIELD
(2) The present invention relates to a fluidic group and related apparatus for drug preparation.
BACKGROUND ART
(3) As is well known, in the field of oncology, the ability to dose chemotherapy drugs precisely and individually for each patient is particularly critical. In fact, chemotherapy drugs used in cancer therapies can generate significant toxic effects, since they have a very narrow window of effective therapeutic concentration (or therapeutic range).
(4) From an operational point of view, there are considerable criticalities in the process of preparation and management of cancer therapies, which is currently still largely based on manual operations by trained operators (e.g. medical staff and/or health professionals). This generates negative consequences in terms of risk for patients and operators, which are difficult to reconcile with an advanced and modern healthcare management.
(5) In particular, the need to manually create chemotherapy drugs (e.g. by pipetting ingredients from respective containers and mixing them together to create the chemotherapy drug) increases the risk of drug contamination and preparation errors. In addition, the operators involved in these operations are exposed to the ingredients and drugs, with potentially negative consequences on their health due to the toxicity of the ingredients.
(6) The main critical elements include the following: poor precision in the formulation of cancer drugs due to human error; need to provide in-depth education and specific training to operators in charge of dosing chemotherapy drugs; difficulty in managing unused quantities of extremely expensive drugs; high occupational risk for operators involved in the preparation of chemotherapy drugs.
DISCLOSURE OF INVENTION
(7) Aim of the present invention is to provide a fluidic group and related apparatus for drug preparation which solve the disadvantages of the known art.
(8) According to the present invention, a fluidic group and related apparatus for drug preparation, as defined in the appended claims, are realised.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a better understanding of the present invention, a preferred, non-limiting embodiment thereof is described below, provided by way of non-limiting example and in a triaxial Cartesian system XYZ with reference to the accompanying drawings, wherein:
(2) FIG. **1** is a schematic perspective view of an apparatus for the preparation of chemotherapeutic drugs, according to an embodiment;
(3) FIG. **1**A is a schematic view from above, in an XY plane, of the apparatus of FIG. **1**;
(4) FIG. **1**B is a schematic side view, in an XZ plane, of the apparatus of FIG. **1**;
(5) FIG. **2** is a schematic perspective view of a fluidic device included in the apparatus of FIG. **1**, according to an embodiment;
(6) FIG. **3** is a schematic exploded view of the fluidic device of FIG. **2**;
(7) FIG. **4** is a schematic exploded view of a detail of the fluidic device of FIG. **2**;
(8) FIGS. **5**A-**5**D are schematic side views, in the XZ plane, of a detail of the apparatus of FIG. **1** in respective operating modes, this detail including the fluidic device of FIG. **2**;
(9) FIG. **6** is a block diagram showing a method of implementing the apparatus of FIG. **1**.
BEST MODE FOR CARRYING OUT THE INVENTION

(10) In particular, the figures are shown with reference to the triaxial Cartesian system XYZ defined by an X axis, a Y axis and a Z axis, orthogonal to each other. Next, a gravity acceleration acting along the Z axis is considered.
(11) In the description below, elements common to the various embodiments of the invention are indicated with the same reference numbers.
(12) FIGS. **1**, **1**A and **1**B show an apparatus for preparation of chemotherapy drugs (hereafter referred to as "apparatus" and indicated by reference number **1**).
(13) The apparatus **1** includes a base **3** configured to be placed on a support (not shown, such as a table, a piece of furniture or a platform), and in particular on a surface (not shown) of the support extending in a horizontal XY plane defined by the axes X and Y. The base **3** has an upper surface **3***a* and a lower surface **3***b* opposite each other along the Z axis, and extending substantially parallel to each other and parallel to the XY plane.
(14) A first rotor **9** and a second rotor **11** are carried by the base **3** and have a substantially flat shape. In particular, the first and the second rotor **9**, **11** have a main extension in the XY plane and have a substantially flat circular shape in the XY plane. The rotors **9**, **11** are arranged superimposed one upon the other, are coaxial to each other and are coupled to the base **3** by means of a central stator **13** fixed to the upper surface **3***a*, extending perpendicularly to said surface **3***a* and arranged centrally to the rotors **9**, **11**. In detail, the stator **13** of rectilinear shape has main extension along a rotation axis **15** parallel to the Z axis and includes for example a rod. The stator **13** has a first lower end (not shown) and a second upper end **13***b*, opposite each other along the Z axis, and is fixed to the base **3** by the first lower end. The rotors **9**, **11** are coupled to the stator **13** in a per se known manner, for example by means of respective bearings (not shown), and are configured to rotate with respect to the stator **13**. In detail, the stator **13** is arranged centrally with respect to the rotors **9**, **11** which rotate, in respective rotation planes (parallel to the XY plane), with respect to the rotation axis **15** (orthogonal to the rotation planes). The rotors **9**, **11** are therefore placed at different levels along the Z axis, and in particular the first (lower) rotor **9** faces the upper surface **3***a* of the base **3** and is at a distance from said upper surface **3***a*, along the Z axis, a distance which is less than the distance shown by the second (upper) rotor **11**. In other words, the first lower rotor **9** is interposed, along the Z axis, between the base **3** and the second upper rotor **11**.
(15) The first rotor **9** includes a plurality of first seats **17** each configured to house a respective fluidic device **21**, better described below. Each fluidic device **21** is coupled to a respective first fluidic container **22** (hereinafter, first container **22**). In particular, each first container **22**—of known type—includes a first body **22***a* (e.g., bottle, vial, ampoule) defining a first internal volume and having a through opening closed by a first cap **22***b* (e.g., of silicone). Each first container **22** is adapted to contain a respective ingredient, and is releasably coupled (e.g., by interlocking) to the respective fluidic device **21** at the cap **22***b*, as better described below. In detail, the first containers **22** and the respective fluidic devices **21** are arranged, angularly equally spaced apart between them, at an outer edge of the first rotor **9**.
(16) The second rotor **11** includes a plurality of second seats **19** configured to house a respective plurality of supports **23**. In particular, each support **23** is releasably coupled to the respective second seat **19**. Each support **23** includes a first surface **23***a* and a second surface **23***b* (FIG. **5**A) opposite to each other along the Z axis, and includes a respective second portion (extending at the second surface **23***b*) shaped so as to releasably fit into the respective second seat **19**, for example by interlocking in order to fix the support **23** to the respective second seat **19**. The second surfaces **23***b* of the supports **23** thus face the second rotor **11**. Each support **23** is releasably coupled to a respective second fluid container **24** (hereinafter, second container **24**) also of known type. In particular, each second container **24** includes a second body **24***a* (e.g., bottle, vial, ampoule) defining a second internal volume and having a through opening closed by a second cap **24***b* (e.g., of silicone). Each second container **24** is adapted to contain a respective chemotherapeutic drug to be administered to a respective patient. Each second container **24** is releasably coupled (e.g., by

interlocking) to the respective support **23** at the cap **22***b*. Thus, each cap **22***b* faces the first surface **23***a* of the respective support **23**. In detail, the second containers **24** and the respective supports **23** are arranged, angularly equally spaced apart between them, at an outer edge of the second rotor **11**.
(17) The second rotor **11** has a larger diameter than the respective diameter of the first rotor **9**.
(18) An operating device **5** is coupled (in particular, fixed) to the upper surface **3***a* of the base **3**. The operating device **5** has a side wall **5***a* facing the rotors **9**, **11**.
(19) The operating device **5** comprises first operating means (not shown), of a type known per se and configured to allow rotation of the rotors **9**, **11** with respect to the stator **13**. According to an embodiment, the first operating means extend starting from the side wall **5***a*. Each rotor **9**, **11** is independently operated such that the rotors **9**, **11** can rotate independently of each other, with different speeds and/or with opposite angular directions (e.g., clockwise and counterclockwise in the top view of FIG. **1**B) and with different angular speeds. For example, the first operating means **27** comprise a first and a second motorised gear (not shown, such as gear wheels). The first motorised gear is configured to engage a first toothing fixed to the first rotor **9** (e.g., fixed to a perimeter surface of the first rotor **9**); the second motorised gear is configured to engage a second toothing fixed to the second rotor **11** (e.g., fixed to a perimeter surface of the second rotor **11**). Alternatively, the first operating means **27** comprise a first and second belt drive configured to cooperate frictionally with the first and, respectively, second rotor **9**, **11** (e.g., with the side surfaces of the rotors **9**, **11**). Alternatively, the first operating means **27** exploit, in a per se known manner, electromagnetic fields to move the rotors **9**, **11**. Alternatively, the first operating means **27** are included in the base **3** and the stator **13** includes respective drive means to actuate the rotors **9**, **11**. Alternatively, the first operating means **27** are based on direct drive technologies.
(20) FIGS. **2-4** show one of the fluidic devices **21** in detail. The fluidic device **21** includes a main body **59** shaped so as to releasably couple with (e.g., fit into) the respective first seat **17**, for example by interlocking so as to releasably fix the fluidic device **21** to the respective first seat **17**.
(21) The fluidic device **21** comprises a fluidic connection means **25** fixed to the main body **59** and adapted to fluidically connect the first container **22** with the second container **24**. The fluidic connection means **25** includes a first hollow needle **60** (hereinafter, also referred to as first needle **60** and having a first and second end **60***a*, **60***b*) and a first tubular element **62** (e.g., of plastic material and having a first and second end **62***a*, **62***b*). The first needle **60** is configured to be coupled to (e.g., inserted into) the second cap **24***b* through the first end **60***a*, as better described below. The first tubular element **62** is flexible, is coupled to the main body **59** through the first end **62***a* and is coupled, through the second end **62***b*, to the second end **60***b* of the first needle **60**. The first tubular element **62** and the first needle **60** are coupled to each other so as to allow a passage of fluid from the main body **59** through the first tubular element **62** and the first needle **60**. In detail, the first tubular element **62** has a first through opening defined between the ends **62***a*, **62***b*, and the first needle **60** has a second through opening defined between the ends **60***a*, **60***b*. The first and second cavity face each other at the second ends **60***b*, **62***b*.
(22) The first needle **60** is supported, in a releasable manner, by a first support portion **152** of the main body **59**, and is movable, as shown below, with respect to the main body **59**. In detail, the first support portion **152** extends starting from, and perpendicularly to, a surface **59′** of the main body **59**, said surface **59′** being facing the containers **22**, **24** when the latter are coupled to the main body **59**. When coupled together, the first support portion **152** supports the first needle **60** so that the second end **60***b* of the first needle **60** is facing the main body **59** (at the surface **59′**), and so that the first end **60***a* of the first needle **60** is facing the first container **22**. Furthermore, in use the fluidic device **21** and the support **23** considered are mutually arranged so that the first needle **60** is facing the second cap **24***b* along the Z axis.
(23) The main body **59** has a housing cavity **59***a* in which a syringe **154** comprising a hollow body **156** is housed (in detail, inclusive). The body **156** defines a cavity **158** having a polygonal (e.g. circular) cross-section. The body **156** includes an end **156***a* having a through opening **160**.

Internally to the cavity **158** there is a plunger (or piston) **150**, movable (e.g., cooperating with sliding) with respect to the body **156** and operable so as to generate a pressure gradient (e.g., at the through opening **160**). In detail, the piston **150** is movable, longitudinally to the body **156**, in the cavity **158** and has a first and second end **150***a*, **150***b* opposite each other. The piston **150** comprises, at the first end **150***a*, a seal element **150′** sliding in a fluid-tight manner in the body **156** (in detail, it slides along an internal surface of the body **156**, facing said cavity) thus defining a first and a second chamber **162***a*, **162***b* having a variable volume in the cavity **158**. The chambers **162***a*, **162***b* are fluid-tightly separated from each other by the seal element **150′** and have respective volumes that are variable as a function the axial position assumed by the piston **150** with respect to the body **156**. In detail, the first chamber **162***a* is in fluid communication with the outside of the syringe **154** via the through opening **160** located at the end **156***a*. For example, the piston **150** is movable from a first position (of compression of the first chamber **162***a*) to a second position (of expansion of the first chamber **162***a*): in the first position, the seal element **150′** is at the end **156***a* and the piston **150** is almost completely housed in the cavity **158**, and therefore the first chamber **162***a* has a volume smaller than the volume of the second chamber **162***b* (in other words, the seal element **150′** has a first distance D.sub.1 from the end **156***a* of the body **156**, and the first chamber **162***a* has a first volume); and in the second position, only the seal element **150′** is in the cavity **158** and the piston **150** is almost completely outside the body **156**, and therefore the first chamber **162***a* has a volume greater than the volume of the second chamber **162***b* (in other words, the seal element **150′** has a second distance D.sub.2 from the end **156***a* of the body **156**, and the first chamber **162***a* has a second volume, the second distance D.sub.2 being greater than the first distance D.sub.1 and the second chamber being greater than the first chamber **162***b*).
(24) The syringe **154** is coupled to the first container **22** by means of a forking element **166** and a first one-way (non-return, or check) valve **168** (hereinafter, first valve **168**), that are also arranged in the housing cavity **59***a*. The first valve **168** has a first and a second end **168***a*, **168***b* opposite to each other, and is operable in an opening state and, alternatively, in a closing state as a function of a first force caused by a first pressure difference between said ends **168***a*, **168***b*. The first pressure difference is a function of a first pressure gradient present at the first valve **168**, which in turn is related to said pressure gradient in the through opening **160**. In fact, the forking element **166** and the first valve **168** are in pneumatic communication with the through opening **160** of the syringe **154**. In a rest condition, the first valve **168** is in the closing state; when, on the other hand, the pressure gradient in the through opening **160** is directed towards the cavity **158** (i.e. towards the inside of the syringe **154**) and has an absolute value greater than a first threshold (in other words, when the first force acting on the first valve **168** is greater than a threshold force), the first valve **168** passes from the closing state to the opening state, allowing a passage of fluid only from the first container **22** towards the syringe **154** (thus preventing a passage of fluid from the syringe **154** towards the first container **22**), and then returns to the closing state when the pressure gradient in the through opening **160** is exhausted. On the other hand, the forking element **166** allows the passage of fluid, coming from the syringe **154**, along a first and a second fluidic path, as better described below.
(25) In detail, the forking element **166** has a first and second end **166***a*, **166***b* opposite each other. At the second end **166***b*, the forking element **166** has a first and second fluidic channel **166′**, **166″** and, at the first end **166***a*, the forking element **166** has a third fluidic channel **166‴**. The fluidic channels **166′**, **166″**, **166‴** have a Y-arrangement between them (i.e. they are joined and fluidically connected to each other in a connection portion). The through opening **160** of the syringe **154** is coupled to (e.g., fixed to, and in fluid communication with) the first end **166***a* of the forking element **166** (in detail, to the third fluidic channel **166‴**). For example, the syringe **154** and the forking element **166** are coupled to each other directly or through a further tubular element (not shown and analogous to the first tubular element **62**).
(26) Further, the first fluidic channel **166′** of the second end **166***b* is coupled to (e.g., fixed to, and

in fluid communication with) the second end **168**_b_ of the first valve **168** through a second tubular element **170** (analogous to the first tubular element **62**). The first end **168**_a_ of the first valve **168** is coupled to (e.g., fixed to, and in fluid communication with) the first cap **22**_b_. In particular, the first end **168**_a_ is coupled to (e.g., fixed to, and in fluid communication with; e.g., directly or through a third tubular element **172** analogous to the first tubular element **62**) an interface means **171** that is releasably couplable to the first container **22**.

(27) The interface means **171** is placed, at least partially, externally to the housing cavity **59**_a_ and includes a second support portion **171**_b_. The second support portion **171**_b_ is fixed to the main body **59**, is external to the housing cavity **59**_a_ and, when the first container **22** is coupled to the fluidic device **21**, is releasably fixed to the second cap **22**_b_ so as to physically support and sustain the first container **22**. The second support portion **171**_b_ includes a second hollow needle **171**_a_ (hereinafter, second needle **171**, analogous to the first needle **60**) configured to be coupled to (e.g., inserted into) the first cap **22**_b_. In detail, when the first container **22** is supported by the second support portion **171**_b_, the second needle **171**_a_ is inserted into the first cap **22**_b_ thereby allowing fluidic communication between the first container **22** and the syringe **154**.

(28) The forking element **166**, the second tubular element **170**, the first valve **168**, the interface means **171** and, if any, the third tubular element **172** define the first fluidic path, which therefore crosses the first and third fluidic channel **166′**, **166′′′** to fluidically connect the first internal volume and the first chamber **162**_a_ with each other.

(29) A second one-way (non-return, or check) valve **176** (hereinafter, second valve **176**) is interposed between the fluidic connection means **25** and the forking element **166**. The second valve **176** has a first and a second end **176**_a_, **176**_b_ opposite each other, and is operable in an opening state and, alternatively, in a closing state as a function of a second force caused by a second pressure difference between said ends **176**_a_, **176**_b_. The second pressure difference is a function of a second pressure gradient present at the second valve **176**, which in turn is related to said pressure gradient in the through opening **160**. In fact, the forking element **166** and the second valve **176** are in pneumatic communication with the through opening **160** of the syringe **154**. In a rest condition, the second valve **176** is in the closing state; when, however, the pressure gradient in the through opening **160** is directed towards the forking element **166** (i.e., towards the outside of the syringe **154**) and has an absolute value greater than a second threshold (e.g., equal to the first threshold; in other words, when the second force acting on the second valve **176** is greater than a further threshold force), the second valve **176** passes from the closing state to the opening state, allowing a passage of fluid only from the syringe **154** towards the first needle **60**, and therefore towards the second container **24** (thus preventing a passage of fluid from the second container **24** towards the syringe **154**), to then return to the closing state when the pressure gradient in the through opening **160** is exhausted.

(30) In detail, the second fluidic channel **166″** of the second end **166**_b_ is coupled to (e.g., fixed to, and in fluid communication with) the second end **176**_b_ of the second valve **176** through a fourth tubular element **178** (analogous to the first tubular element **62**). The first end **176**_a_ of the second valve **176** is coupled to (e.g., fixed to, and in fluid communication with) the first end **62**_a_ of the first tubular element **62**. The second valve **176** and the fourth tubular element **178** are also included in the housing cavity **59**_a_.

(31) The forking element **166**, the fourth tubular element **178**, the second valve **176** and the fluidic connection means **25** define the second fluidic path, which therefore crosses the second and the third fluidic channels **166″**, **166′′′** to fluidically connect the first chamber **162**_a_ and the second internal volume with each other.

(32) More in detail, the fluidic device **21** comprises the following elements, which are assembled together (e.g. by gluing) so as to form a single group adapted to transfer liquids without leakage: the interface means **171**: this is a ventilated spike, adapted to pierce the first cap **22**_b_ (i.e. an elastomer cap normally present in drug bottles) and to withdraw the ingredient from the first

selected container **22**, with compensation of the internal pressure of the first container **22** (the spike is provided with a double lumen: one through which the ingredient is withdrawn from the first container **22**, the other allowing the entry into the first container **22** of air in equal volume, in order to compensate for the pressure inside the first body **22***a*); the first valve **168** and the second valve **176**: these are one-way valves, each allowing the ingredient to pass only in the direction indicated by the respective arrow in FIG. **4**; the forking element **166**: it is a "Y" fitting, provided with a luer-lock connector for connection to the syringe **154**; the syringe **154**: this is a luer-lock syringe; the first needle **60**: it is a male, non-valved luer-slip connector, suitable for connection to female luer-slip or luer-lock connectors, whether or not valved; and the first tubular element **62**, the second tubular element **170**, the third tubular element **172** and the fourth tubular element **178**: these are respective connection tubes between the various elements mentioned above, as described in more detail above.

(33) With reference to FIGS. **1**B, **5**A, the operating device **5** further comprises second operating means **70**, configured to move the first needle **60** by displacing it upwards (thus towards the rotor **11** along the Z axis) so as to couple the first needle **60** of the fluidic device **21** with the second cap **24***b* of the second container **24** considered. In particular, the second operating means **70** extend starting from the side wall **5***a*, and include a first translation means **72** and a first gripping means **74**, operatively coupled to each other.

(34) The first gripping means **74** is configured to couple to the first needle **60**, allowing for the aforesaid movement. In particular, the first gripping means **74** includes two arms **74′** having respective end portions **74″**. For example, the arms **74′** have main extension parallel to the X axis, are arranged so that the first needle **60** is placed, parallel to the Y axis, between these end portions **74″**, and the arms **74′** can be operated so that they are closed around the needle **60** and allow it to be seized. In a first operating condition, the end portions **74″** are decoupled from each other (they are arranged with each other along the Y axis at a first distance D.sub.1); and in a second operating condition, the end portions **74″** are coupled with each other (they are arranged with each other along the Y axis at a second distance D.sub.2 less than the first distance D.sub.1). In particular, in the second operating condition, the end portions **74″** abut with the first needle **60**, which is therefore integral with the end portions **74″** by friction and/or interlocking with the latter. The arms **74′** are operated in a manner per se known (e.g., by means of gears, by means of a motor, e.g., a stepper type, or by means of at least one translation means defining a respective path parallel to the Y axis and analogous to the first translation means **72** described below).

(35) The first translation means **72** is configured to displace the first needle **60** with respect to the main body **59** towards the second rotor **11**. The first translation means **72** comprises a first guide element **73** and a first movable element **75**, which are movable between them (e.g., cooperating with each other by sliding). In particular, the first guide element **73** is fixed to the side wall **5***a* and has main extension parallel to the Z axis. In other words, the first guide element **73** defines a first path along which the first movable element **75** is constrained to move, said first path being linear and parallel to the Z axis. In detail, the first guide element **73** extends, along the Z axis, from the level of the first needle **60** to the level of the cap **24***b*.

(36) The first movable element **75** is moved in the first guide element **73** by means of first motor means (not shown). In particular, the first motor means are fixed to the side wall **5***a* or to the first guide element **73**, and are coupled to the first movable element **75** so as to allow it to be displaced in the first guide element **73** from a rest position to an activation position. For example, the first motor means include a first stepper motor, or linear actuators based on geared motors, or actuators of the pneumatic or hydraulic type.

(37) The subsequent actions of the first gripping means **74** and the first translation means **72** thus make it possible to seize the first needle **60** arranged in the rest position and, subsequently, to displace it parallel to the Z axis until bringing it into the activation position in which the first needle **60** is inserted into the second cap **24***b*, thus allowing a fluidic connection between the first

container **22** and the second container **24**.
(38) The operating device **5** further comprises third operating means **80** (or pumping means), configured to operate the fluidic device **21** so as to allow a passage of fluid between the first container **22** and the first needle **60**. In particular, the third operating means **80** are controllable to operate the piston **150** in order to pump fluid from the first container **22** to the second container **24**.
(39) For example, the third operating means **80** extend starting from the side wall **5**_a_, and include a second translation means **272** and a second gripping means **274**, operatively coupled to each other.
(40) The second gripping means **274** is configured to couple to the piston **150** at the second end **150**_b_. In particular, the second gripping means **274** includes two respective arms **274′** having respective end portions **274″**. For example, the arms **274′** have main extension parallel to the X axis, are available such that the second end **150**_b_ of the piston **150** is available, parallel to the Y axis, between these end portions **274″**, and the arms **274′** can be operated so that they are closed around the piston **150** and allow it to be seized. In a first operating condition, the end portions **274″** are decoupled from each other (e.g., they are arranged with each other along the Y axis at a third distance D.sub.3); and in a second operating condition, the end portions **274″** are coupled with each other (e.g., they are arranged with each other along the Y axis at a fourth distance D.sub.4 less than the third distance D.sub.3). In particular, in the second operating condition, the end portions **274″** abut with the second end **150**_b_ of the piston **150**, which is therefore integral with the end portions **274″** by friction and/or interlocking with the latter. The arms **274′** are operated in a manner per se known (e.g., by means of gears, by means of a motor, e.g., a stepper type, or by means of at least one translation means defining a respective path parallel to the Y axis and analogous to the first translation means **72**).
(41) The second translation means **272** is configured to displace the piston **150** with respect to the main body **59**; this displacement of the piston **150** occurs, alternately, towards the side wall **5**_a_ and towards the main body **59**. The second translation means **272** comprises a second guide element **273** and a second movable element **275**, which are movable between them (e.g., cooperating with each other by sliding). In particular, the second guide element **273** is fixed to the side wall **5**_a_ and has main extension parallel to the X axis. In other words, the second guide element **273** defines a second path along which the second movable element **275** is constrained to move, said second path being linear and parallel to the X axis.
(42) The second movable element **275** is moved in the second guide element **273** by means of second motor means (not shown). In particular, the second operating means are fixed to the side wall **5**_a_ or to the second guide element **273**, and are coupled to the second movable element **275** so as to allow it to be displaced in the second guide element **273** from a first position (e.g., in which the piston **150** is in the compression position of the first chamber **162**_a_) to a second position (e.g., in which the piston **150** is in the expansion position of the first chamber **162**_a_), and vice versa. For example, the third motor means include a third stepper motor, or linear actuators based on geared motors, or actuators of the pneumatic or hydraulic type.
(43) The subsequent actions of the second gripping means **274** and the second translation means **272** thus make it possible to seize the piston **150** arranged in the compression position of the first chamber **162**_a_ and, subsequently, to displace it parallel to the X axis until it is brought into the expansion position of the first chamber **162**_a_, and vice versa. This makes it possible to create said pressure gradients, to actuate the valves **168**, **176** and to transfer part of the ingredient from the first container **22** to the second container **24**, as better described below.
(44) A control unit **83** (such as a processor, ASIC, PCB, or dedicated controller) is also operatively coupled to the apparatus **1** and is configured to command the apparatus **1**, and in particular to command the first operating means **27**, the second operating means **70**, and the third operating means **80**. In particular, the apparatus **1** includes the control unit **83** (e.g., the control unit **83** is included in the base **3** or in the operating device **5**).
(45) Optionally, the apparatus **1** further comprises first interface means **85′**, for example including

at least one between a display device (such as a screen, also of the touch screen type) and a plurality of buttons. For example, the first interface means **85′** are included in the base **3** or in the operating device **5**. In addition, the apparatus **1** may also be operatively coupled (e.g., wired, via the Internet or electromagnetically) to second interface means **85″** (not shown), such as external electronic devices (e.g., computer, smartphone, keyboard or mouse). In particular, the second interface means **85″** may include a printing apparatus, such as a label printer. In particular, the first interface means **85′** and the second interface means **85″** are operatively coupled to the control unit **83**.

(46) With reference to FIGS. **5**A-**5**D and **6**, a method for operating **100** the apparatus **1**, implemented via the control unit **83**, is described.

(47) The operating method **100** starts at step **102**. Initially, the supports **23**, the fluidic devices **21** and the containers **22**, **24** are not coupled to the apparatus **1**. At step **102**, the device **1** is turned on and acquires prescription data which are entered, either by the operator or in an automated manner, via the first interface means **85′** and/or second interface means **85″** (e.g., a server of a hospital facility comprising an archive with patient data). The prescription data are indicative of the prescription to be followed, i.e. the chemotherapy drug to be prepared for the patient under consideration. In particular, the prescription data include: patient's personal data (name, surname, date of birth, any additional identifying information); active substance (or ingredient) of the chemotherapeutic drug to be prepared; quantity of active substance (or ingredient) prescribed; trade name and size (e.g. expressed in mL) of the second container **24** adapted to contain the chemotherapeutic drug; type of second container **24** (e.g. bag, syringe, elastomer); any substance already present in the second container **24** (e.g. in the case of bags) and relative volume; and delivery times.

(48) For example, this quantity of active substance of the chemotherapy drug is delivered in weight W.sub.pa (e.g., in mg), and is converted into a corresponding weight W.sub.fc of commercial chemotherapy drug (e.g., in mg) using the following relationship:

(49) $P_{\mathrm{fc}} = \frac{P_{pa}}{c_{\mathrm{fc}}}$ .Math. d_{fc} where W.sub.fc is the weight (in mg) of the chemotherapy drug to be dosed and administered to the patient under consideration, W.sub.pa is the weight (in mg) of the active substance (or ingredient), d.sub.fc is a density (in mg/mL) of the chemotherapy drug and cfc is a concentration of the active substance (or ingredient) in the chemotherapy drug.

(50) In step **102**, a set-up of the apparatus **1** is also performed, for example performed in a controlled atmosphere (e.g., under a hood) and including: disinfection and cleaning of apparatus **1**, of the supports **23**, of the fluidic devices **21** and of the containers **22**, **24**; and assembly of the supports **23**, of the fluidic devices **21** and of the containers **22**, **24** to the apparatus **1**.

(51) In particular, said assembly of the supports **23**, of the fluidic devices **21** and of the containers **22**, **24** to the apparatus **1** is performed by the operator or in an automated manner, and allows loading all the materials and/or objects necessary for the preparation of the specific chemotherapeutic drug. The supports **23**, the fluidic devices **21** and the containers **22**, **24** are identified by reading respective identification codes associated therewith. In particular, the supports **23**, the fluidic devices **21** and the containers **22**, **24** have either respective labels glued thereto and including respective barcodes, or respective radio frequency identification (“RFID”) labels, suitable for identifying them. The fluidic devices **21** and the first containers **22** are releasably fixed to each other, and then the fluidic devices **21** carrying the first containers **22** are coupled to the first rotor **9**, as previously described. Similarly, the supports **23** and the second containers **24** are releasably fixed to each other, and then the supports **23** carrying the second containers **24** are coupled to the second rotor **11**, as previously described.

(52) This is followed, at a step **104**, by a positioning of the rotors **9**, **11**, shown in FIG. **5**A. In detail, the second rotor **11** rotates, by means of the first operating means **27**, so as to bring the second selected container **24** closer to the operating device **5**, said second container **24** being adapted to contain the chemotherapeutic drug of the patient under consideration. Furthermore, the

first rotor **9** rotates, by means of the first operating means **27**, so as to bring the selected first container **22** closer to the operating means **5**, said first container **22** containing the ingredient to be inserted into the second container **24** to make the chemotherapeutic drug for the patient under consideration. The selected containers **22**, **24** are therefore overlapping and at least partially aligned with each other parallel to the Z axis and face the side wall **5***a* of the operating device **5**.
(53) As shown in FIG. **5**B, at a step **106**, consecutive to step **104**, the second operating means **70** couple the first needle **60** of the selected fluidic device **21** with the second selected container **24**, said first needle **60** is used to achieve fluidic communication between the containers **22**, **24** that have been selected.
(54) As shown in FIG. **5**C, at a step **108**, consecutive to step **106**, the third operating means **80** are coupled to the fluidic device **21**, as further described below.
(55) As shown in FIG. **5**D, at a step **110**, consecutive to step **108**, fluid is transferred from the first container **22** to the second container **24**, as better described below.
(56) At a step **112**, consecutive to step **110**, the third operating means **80** are decoupled from the fluidic device **21**, performing a reverse movement to that described with reference to step **108** and, hereinafter, FIG. **5**C.
(57) At a step **114**, consecutive to step **112**, the second operating means **70** displace the first needle **60** downwards (i.e. in the direction of the base **3** along the Z axis) and decouple the first needle **60** from the second container **24**, interrupting fluidic communication between the containers **22**, **24**. In other words, a reverse movement to that described with reference to step **106** is made.
(58) At step **116**, consecutive to step **114**, a condition on the composition of the chemotherapeutic drug to be prepared is verified. In particular, it is determined whether other ingredients (located in respective first containers **22** other than the previously considered first container **22**) need to be added to the second container **24** to obtain the chemotherapeutic drug.
(59) If further ingredients are needed for the preparation of the chemotherapeutic drug (output "S" from step **116**), the positioning of the rotors **9**, **11** is performed again, thus returning to step **104**. In particular, the first rotor **9** is operated to select the first container **22** of interest.
(60) If no further ingredients are needed for the preparation of the considered chemotherapeutic drug (output "N" from step **116**), the operating method **100** ends at a step **118**, wherein, for example, the device **1** is turned off or will be awaiting a further order for a different preparation.
(61) Furthermore, at step **118**, a disassembly of the supports **23**, of the fluidic devices **21** and of the containers **22**, **24** from the apparatus **1** may be performed by the operator or in an automated manner, in order to prepare the apparatus **1** to perform step **102** again.
(62) Furthermore, at step **118**, an identification label for the second container **24** may be generated, for example, by means of said printing apparatus (included in the second interface means **85**″ and operatively coupled to the control unit **83**). In particular, such a label may include, for example, a barcode label or an RFID label. Such a label is adapted to be coupled (e.g., glued or fixed) to the second container **24**, and is indicative of data such as the nature of the chemotherapeutic drug contained in the second container **24** (e.g., chemical composition and/or active substance contained) and/or an identifier of the patient to whom said chemotherapeutic drug is to be administered.
(63) In more detail and with reference to FIGS. **5**B-**5**D, the operation of the third operating means **80** (shown here in their respective operating modes) is described.
(64) In FIG. **5**B, the third operating means **80** are in a position decoupled from the piston **150**, and are therefore in a rest position.
(65) In FIG. **5**C, the third operating means **80** are coupled to the piston **150** (step **108**). In particular, the second translation means **272** translates along the X axis the second gripping means **274** so that the second end **150***b* of the piston **150** is interposed, parallel to the Y axis, between the end portions **274**″. Subsequently, the second gripping means **274** is coupled to the piston **150**, placing the end portions **274**″ in the second operating condition so that they are releasably fixed to

the second end **150**_b_ of the piston **150**.

(66) In FIG. **5**D, the third operating means **80** cause part of the ingredient to be transferred from the first container **22** to the syringe **154**. In detail, the second translation means **272** translates along the X axis the second gripping means **274** and the piston **150**, thus moving the first end **150**_a_ of the piston **150** away from the end **156**_a_ of the body **156**. In other words, it passes from the compression position of the first chamber **162**_a_ to the extension position of the first chamber **162**_a_. This generates the first pressure gradient (and thus lower pressure at the second ends **168**_b_, **176**_b_ than at the first ends **168**_a_, **176**_a_) which brings, when the first force is greater than the threshold force, the first valve **168** into the opening state (while the second valve **176** remains in the closing state). Furthermore, the first pressure gradient causes, together with the force of gravity, the transfer of the ingredient from the first container **22** to the first chamber **162**_a_ of the syringe **154**.

(67) Similarly to the foregoing, a movement (not shown) of the third operating means **80** opposite to that shown in FIG. **5**D causes the transfer of part of the ingredient from the syringe **154** to the second container **24**. In detail, with reference to the configuration shown in FIG. **5**D, the second translation means **272** translates again along the X axis the second gripping means **274** and the piston **150** in the opposite direction with respect to the moment described with reference to FIG. **5**D, thus bringing the first end **150**_a_ of the piston **150** closer to the end **156**_a_ of the body **156**. In other words, there is a passage from the expansion position of the first chamber **162**_a_ to the compression position of the first chamber **162**_a_. This generates the second pressure gradient (and thus greater pressure at the second ends **168**_b_, **176**_b_ than at the first ends **168**_a_, **176**_a_) which brings, when the second force is greater than the further threshold force, the second valve **176** into the opening state (while the first valve **168** remains in the closing state). In addition, the second pressure gradient causes the transfer of the ingredient from the first chamber **162**_a_ of the syringe **154** to the second container **24**.

(68) In conjunction with each other, the expansion and compression movements of the first chamber **162**_a_ operated by the third operating means **80** allow the transfer of the ingredient from the first container **22** to the second container **24** (step **110**).

(69) From an examination of the characteristics of the invention realised according to the present invention, the advantages that it allows obtaining are evident.

(70) The apparatus **1** is an automated system for the preparation of anticancer drugs, capable of overcoming the critical elements connected with the manual preparation thereof and of ensuring a substantial increase in the quality of cancer therapies. In fact, the apparatus **1** allows a significant increase in formulation accuracy, a significant reduction of chemical contamination of the environments intended for the preparation of such drugs and a drastic reduction of the professional risk for the healthcare workers assigned thereto.

(71) In addition, the apparatus **1** implements, via the control unit **83**, a computerised procedure for managing chemotherapy preparations, for use for example by hospital pharmacies. This computerised procedure can be interfaced with a hospital information system (if any) through standard communication protocols.

(72) The apparatus **1** is an automated table-top device, which can be easily inserted into a laminar flow hood for chemotherapy. For example, the apparatus **1** may be operated under a vertical laminar flow hood of ISO class 5, which in turn is placed in an ISO class 8 air-controlled environment.

(73) Advantageously, the communication between the control unit **83** and the second interface means **85**″ takes place in wireless mode, in order to eliminate any wired connection between a sterile interior of the hood in which the apparatus **1** is positioned and the external environment.

(74) In particular, the apparatus **1** allows the adoption of a closed circuit fluidic system able to guarantee sterility during the transfer of the chemotherapeutic drug from the first container **22** to the second container **24**, and the absence of contamination of the surrounding environment.

(75) In particular, the fluidic device **21** is made of low-cost plastic materials for medical use, such

as polystyrene, polycarbonate, polyurethane or PVC and has a structure that is easy to manufacture. This ensures reduced manufacturing costs and allows for single use, of the disposable type. In other words, the fluidic device **21** is coupled only once to the respective first container **22**, and after the transfer of the respective ingredient it is thrown away. This significantly reduces the risk of contamination for patients and healthcare workers.
(76) The fluidic device **21** is a standardised interface allowing the coupling of different types of first containers **22** to the first rotor **9**.
(77) In addition, the chemotherapy drug is dosed directly into the second containers **24**, which are the final containers intended for administration to the patient. This avoids the need for further and subsequent handling, which could be a source of further contamination.
(78) Each second container **24** is identified by a code (e.g. QR) with the details of the chemotherapy drug preparation and of the patient for whom it is intended. This information is therefore verifiable at the time of administration in order to prevent possible accidental exchange of drugs.
(79) The main advantages obtainable by means of apparatus **1** are: increase in the accuracy of chemotherapy drug dosing; increase in the sterility level of injectable preparations; reduction of cytostatic contamination of the surrounding environment (laboratory, pharmacy, hospital); more safety for hospital pharmacy workers (reduced needlestick injuries, reduction of hand fatigue pathologies, etc.); more patient safety thanks to the reduction of the risk of accidental drug and/or patient exchanges, or incorrect dosing; optimisation of the use of cytostatic drugs, with a reduction in the significant costs related to discarding unused cytostatic drugs; and simplification of the drug preparation process, reducing machine set-up times.
(80) In addition, the fluidic group **21** allows the connection, commanded by the second operating means **70** of the apparatus **1**, between the first container **22** (i.e. the bottle with the selected ingredient) and the second container **24** (or final container of the drug, i.e. bag, syringe, etc.), aimed at transferring the ingredient itself in the predetermined quantity controlled by the apparatus **1**.
(81) As regards the first container **22**, this connection with the fluidic group **21** is permanent and is consequent to the perforation of the first cap **22***b* by the interface means **171** and to the engagement of the metal capsule of the first cap **22***b* by a ring nut (e.g., of plastic) of the interface means **171**.
(82) As regards the second container **24**, this connection with the fluidic unit **21** is made by means of the second operating means **70**, which provide for the insertion of the first needle **60** (i.e., a non-valved male luer-slip connector) into a female Luer-lock connector (valved) previously coupled to the second container **24** (in detail, inserted into the second cap **24***b*).
(83) In particular, the first needle **60** makes it possible to use the fluidic device **21** in the automated apparatus **1**.
(84) It is clear that changes and variations can be made to the invention described and shown herein without departing from the protection scope of the present invention, as defined in the appended claims.
(85) In particular, the first operating means **27** may also be arranged in the base **3** instead of in the operating means **5**.
(86) In addition, the second rotor **11** may be replaced by a support structure (not shown) fixed to the base **3** (therefore not rotatable) in the event that only a second container **24** can be coupled to the apparatus **1**. In such a case, after the ingredients have been added to the second container **24** to create the chemotherapeutic drug in a manner similar to that described above, said second container **24** is removed and replaced, automatically or manually, with a further second container **24** to proceed to a further preparation of chemotherapeutic drug.
(87) The apparatus **1** may also be used in the preparation of drugs other than chemotherapeutic drugs, for example injectable drugs such as monoclonal antibodies, cytotoxics, hormones and hormone antagonists, antimitotics, alkylating agents, antibiotics.

Claims

1. A fluidic group (**21**) couplable to a first container (**24**) defining a first internal volume adapted to contain a drug, and to a second container (**22**) defining a second internal volume adapted to contain an ingredient for the preparation of said drug, the fluidic group comprising a first body (**59**) defining a first cavity (**59***a*) and including: a syringe (**154**) comprising a second body (**156**) defining a second cavity (**158**) housing a piston (**150**) movable in the second body (**156**) and defining, with the second body (**156**), a chamber (**162***a*) having a variable volume as a function of a position of the piston (**150**) in the second cavity (**158**); a first coupling group (**171**, **172**) extending partially outside the first cavity (**59***a*) and fixable in a releasable manner to the second container (**22**) so as to be in fluid communication with the second internal volume; a second coupling group (**25**) extending partially outside the first cavity (**59***a*) and fixable in a releasable manner to the first container (**24**) so as to be in fluid communication with the first internal volume; a forking element (**166**) having a first (**166′**) and a second (**166″**) inlet arranged in fluid communication with the first coupling group (**171**, **172**) and, respectively, with the second coupling group (**25**), and furthermore having a third inlet (**166‴**) in fluid communication with said chamber (**162***a*), the third inlet further communicating with the first and second inlets so as to create a first fluidic path between the second internal volume and said chamber (**162***a*) and a second fluidic path between said chamber (**162***a*) and the first internal volume; a first valve (**168**) of one-way type, interposed along the first fluidic path between the first coupling group (**171**, **172**) and the forking element (**166**) and controllable, according to a first pressure gradient caused by a movement of the piston (**150**) in the cavity (**158**), to allow the passage of the ingredient from the second container (**22**) to the chamber (**162***a*) of the syringe (**154**); and a second valve (**176**) of one-way type, interposed along the second fluidic path between the second coupling group (**25**) and the forking element (**166**) and controllable, as a function of a second pressure gradient opposite the first pressure gradient, to allow the passage, from said chamber (**162***a*) to the first container (**24**), of the ingredient transferred from the second container (**22**) into the chamber (**162***a*) of the syringe (**154**).
2. The fluidic group according to claim 1, wherein the second body (**156**) has a first (**156***a*) and a second (**156***b*) end portion opposite each other, the chamber (**162***a*) being fluidically connected to the third inlet (**166‴**) at said first end portion (**156***a*), wherein the piston (**150**) comprises a seal element (**150′**) defining with the second body (**156**) said chamber (**162***a*) and sliding in a fluid-tight manner in the second body (**156**) between a first position and a second position relative to the second body (**156**), wherein, when the seal element (**150′**) is in the first position, the seal element (**150′**) has a first distance from the first end portion (**156***a*) of the second body (**156**) and the chamber (**162***a*) has a first volume, and wherein, when the seal element (**150′**) is in the second position, the seal element (**150′**) has a second distance from the first end portion (**156***a*) of the second body (**156**) and the chamber (**162***a*) has a second volume, the second distance being greater than the first distance and the second volume being greater than the first volume.
3. The fluidic group according to claim 1, wherein pumping means (**80**) are couplable in a releasable manner to the piston (**150**) and are operable to generate the first pressure gradient or the second pressure gradient by moving the piston (**150**) in the second cavity (**158**).
4. The fluidic group according to claim 2, wherein the first valve (**168**) is operable to pass from a first closing state to a first opening state and the second valve (**176**) is operable to pass from a second closing state to a second opening state, and wherein, when the seal element (**150′**) is moved by the pumping means (**80**) from the first position to the second position, the first pressure gradient exerts a force such that the first valve (**168**) is in the first opening state and the second valve (**176**) is in the second closing state, and when the seal element (**150′**) is moved by the pumping means (**80**) from the second position to the first position, the second pressure gradient exerts a further force such that the first valve (**168**) is in the first closing state and the second valve (**176**) is in the

second opening state.

5. The fluidic group according to claim 1, wherein the second coupling group (**25**) comprises a first flexible tubular element (**62**) and a first hollow needle (**60**) fluidically coupled with the second valve (**176**) through said first tubular element (**62**), the first hollow needle (**60**) being movable with respect to the first body (**59**) and insertable in a first cap (**24***b*) of the first container (**24**) so as to allow said fluid communication between said chamber (**162***a*) of the syringe (**154**) and the first internal volume.

6. The fluidic group according to claim 5, wherein the first body (**59**) has a first surface (**59′**) faceable to the second container (**22**), wherein the first hollow needle (**60**) is supported in a releasable manner by a first support portion (**152**) of the first body (**59**) extending starting from, and perpendicular to, said first surface (**59′**), wherein the first hollow needle (**60**) has a first (**60***a*) and a second (**60***b*) end opposite to each other, the second end (**60***b*) being fixed to the first tubular element (**62**), and wherein the first hollow needle (**60**) is supported by the first support portion (**152**) so that the second end (**60***b*) is faceable to the first surface (**59′**) and the first end (**60***a*) is faceable to the first cap (**24***b*) of the first container (**24**).

7. The fluidic group according to claim 1, wherein the first coupling group (**171**, **172**) comprises a second support portion (**171***b*) fixed to the first body (**59**), extending outside the first cavity (**59***a*) and configured to support the second container (**22**), the second support portion (**171***b*) including a second hollow needle (**171***a*) in fluid communication with the first inlet (**166′**) through the first valve (**168**), the second hollow needle (**171***a*) being insertable in a second cap (**22***b*) of the second container (**22**) so as to allow said fluid communication between the second internal volume and said chamber (**162***a*) of the syringe (**154**).

8. An apparatus (**1**) for drug preparation starting from at least one ingredient, the apparatus (**1**) being characterized in that it comprises: a first support element (**11**), configured to support at least one first container (**24**) adapted to contain the drug and arrangeable, in a releasable manner, on the first support element (**11**); at least one fluidic group (**21**) according to claim 1; a second support element (**9**), supporting the at least one fluidic group (**21**), said at least one fluidic group (**21**) being arranged, in a releasable manner, on the second support element (**9**) and being couplable in a releasable manner with a respective second container (**22**) adapted to contain a respective said ingredient; a control unit (**83**); connection means (**70**), couplable to the first container (**24**) and to the second container (**22**) and operatively coupled to the control unit (**83**), said connection means (**70**) being controllable by the control unit (**83**) so as to couple the second coupling group (**25**) with the first container (**24**) to fluidically connect the second container (**22**) to the first container (**24**); and pumping means (**80**), coupled to the fluidic group (**21**) and operatively coupled to the control unit (**83**), said pumping means (**80**) being controllable by the control unit (**83**) so as to transfer, at least partially, the ingredient from the second container (**22**) into the first container (**24**).

9. The apparatus according to claim 8, further comprising a base (**3**) having a second surface (**3***a*) to which the first support element (**11**) and the second support element (**9**) are coupled, the first support element (**11**) and the second support element (**9**) extending parallel to said second surface (**3***a*) and being arranged coaxially to each other relative to a rotation axis (**15**) perpendicular to the second surface (**3***a*), and the second support element (**9**) being axially interposed, relative to the rotation axis (**15**), between the first support element (**11**) and the second surface (**3***a*), and being rotatable relative to the base (**3**) about the rotation axis (**15**).

10. The apparatus according to claim 9, wherein the second support element (**9**) is rotatable relative to the base (**3**) about the rotation axis (**15**) and is configured to support, in a releasable manner, the second container (**22**) through the fluidic group (**21**), and at least a further second container (**22**) through a further fluidic group (**21**).

11. The apparatus according to claim 9, wherein the first support element (**11**) is rotatable relative to the base (**3**) about the rotation axis (**15**) and is configured to support, in a releasable manner, the first container (**24**) and at least a further first container (**24**).

12. The apparatus according to claim 10, further comprising first operating means (**27**) controllable by the control unit (**83**) and configured to rotate, independently of each other, the first support element (**11**) and the second support element (**9**) relative to the base (**3**) about the rotation axis (**15**), so that a first selected container (**24**) and a second selected container (**22**) are arrangeable in respective angular positions so that they are facing the connection means (**70**) and the pumping means (**80**).
13. The apparatus according to claim 9, wherein the connection means (**70**) comprise: a first guide element (**73**), fixed to the base (**3**) and defining a first path parallel to a first axis (Z); a first movable element (**75**), coupled with the first guide element (**73**) so as to move along said first path; and a first gripping means (**74**), fixed to the first movable element (**75**) and controllable by the control unit (**83**) to couple, in a releasable manner, with the second coupling group (**25**), wherein the connection means (**70**) are controllable by the control unit (**83**) to couple, through the first gripping means (**74**), with the second coupling group (**25**) and, subsequently, to move the second coupling group (**25**) along the first path so as to couple the second coupling group (**25**) with the first container (**24**) creating a connection with passage of fluid.
14. The apparatus according to claim 13, wherein the pumping means (**80**) include: a second guide element (**273**), fixed to the base (**3**) and defining a second path parallel to a second axis (X) orthogonal to the first axis (Z); a second movable element (**275**), coupled with the second guide element (**273**) so as to move along said second path; and a second gripping means (**274**), fixed to the second movable element (**275**) and controllable by the control unit (**83**) to couple, in a releasable manner, with the piston (**150**), wherein the pumping means (**80**) are controllable by the control unit (**83**) to couple, by means of the second gripping means (**274**), with the piston (**150**) and, subsequently, to move the second movable element (**275**) relative to the second guide element (**273**) along the second path, moving the piston (**150**) in the second cavity (**158**) and generating the first pressure gradient or the second pressure gradient.
15. The apparatus according to claim 14, wherein the second body (**156**) has a first (**156***a*) and a second (**156***b*) end portion opposite each other, the chamber (**162***a*) being fluidically connected to the third inlet (**166‴**) at said first end portion (**156***a*), wherein the piston (**150**) comprises a seal element (**150′**) defining with the second body (**156**) said chamber (**162***a*) and sliding in a fluid-tight manner in the second body (**156**) between a first position and a second position relative to the second body (**156**), wherein, when the seal element (**150′**) is in the first position, the seal element (**150′**) has a first distance from the first end portion (**156***a*) of the second body (**156**) and the chamber (**162***a*) has a first volume, wherein, when the seal element (**150′**) is in the second position, the seal element (**150′**) has a second distance from the first end portion (**156***a*) of the second body (**156**) and the chamber (**162***a*) has a second volume, the second distance being greater than the first distance and the second volume being greater than the first volume, wherein the first valve (**168**) is operable to pass from a first closing state to a first opening state and the second valve (**176**) is operable to pass from a second closing state to a second opening state, wherein the pumping means (**80**) are controllable by the control unit (**83**) to move the seal element (**150′**) from the first position to the second position, the first pressure gradient exerting a force such that the first valve (**168**) is in the first opening state and the second valve (**176**) is in the second closing state to allow passage of the ingredient from the second container (**22**) into the chamber (**162***a*) of the syringe (**154**), and wherein the pumping means (**80**) are controllable by the control unit (**83**) to move the seal element (**150′**) from the second position to the first position, the second pressure gradient exerting a further force such that the first valve (**168**) is in the first closing state and the second valve (**176**) is in the second opening state to allow the passage, from the chamber (**162***a*) of the syringe (**154**) to the first container (**24**), of the ingredient transferred from the second container (**22**) to the chamber (**162***a*).
16. The apparatus according claim 8, wherein the control unit (**83**) is operatively coupled to a printing apparatus (**85″**) controllable by the control unit (**83**) to generate a label adapted to be coupled with the first container (**24**) and identifying said drug.